S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

(An Autonomous Institute, Affiliated to RTMNU, Nagpur)

DEPARTMENT OF EMERGING TECHNOLOGIES CSE (AI&ML) and CSE (DS)

"Become an excellent center for Emerging Technologies in Computer Science to create competent professionals"

PriceRadar — Real-Time Price Intelligence System

PROJECT TITLE SEMINAR

Group No. 8

Session 2026-27 (ODD)

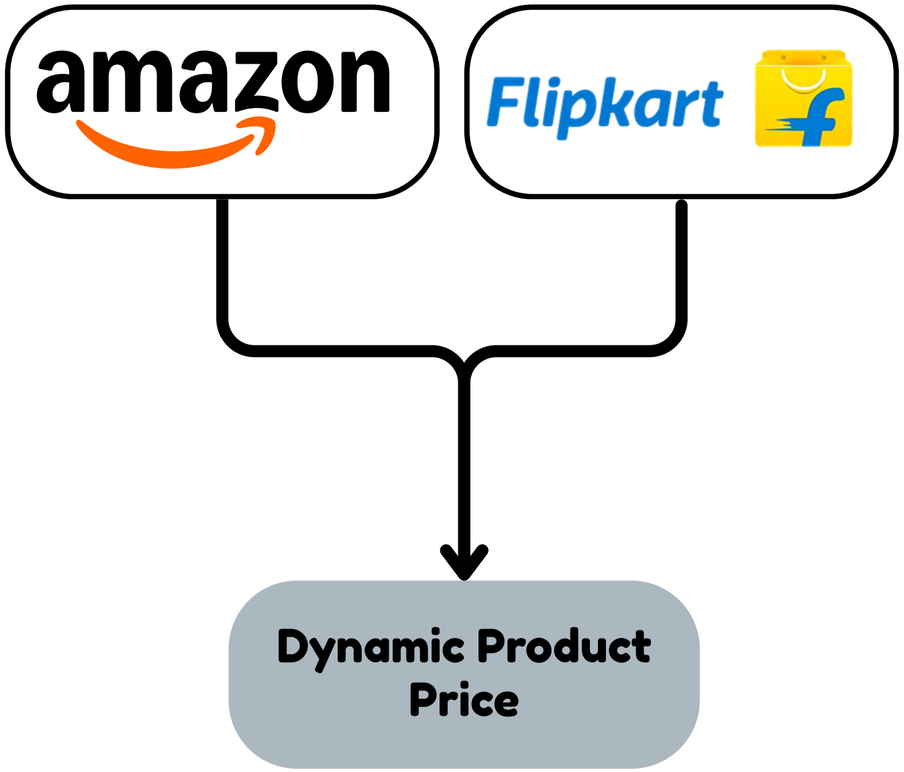
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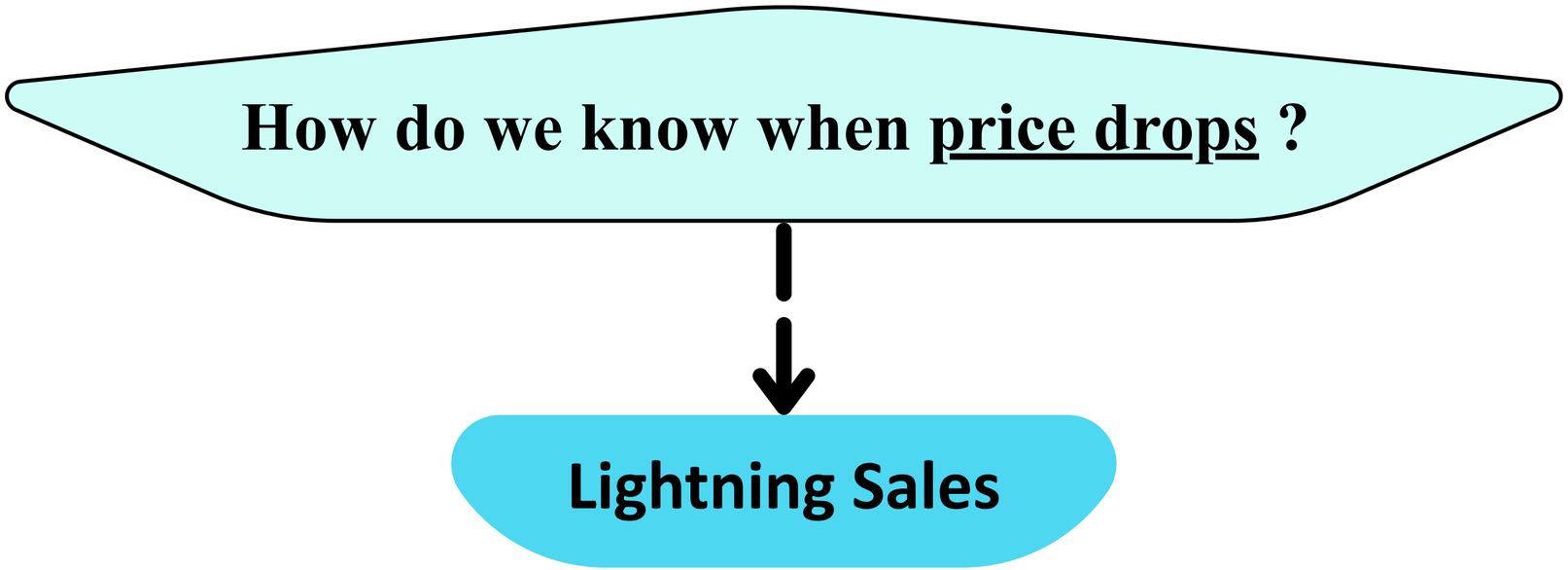
1. **Literature Survey**
2. **Problem Definition**
3. **Objective & Scope**
4. **Methodology**
5. **Plan of Action**
6. **Software & Hardware Requirement**
7. **References**

Price Tracker for the ecommerce platforms like Amazon and flipkart

Problem Definition

We all Shop Online on **Amazon** or **Flipkart** . We all know that the prices on this platform are not fixed they changed multiple times it may drop up to **30-40 %** of it's actual MRP. The problem is " **How do we know when price drops ?** " we can't keep refrash the product page multiple times. So either we ovrer pay or just miss the deal completly .



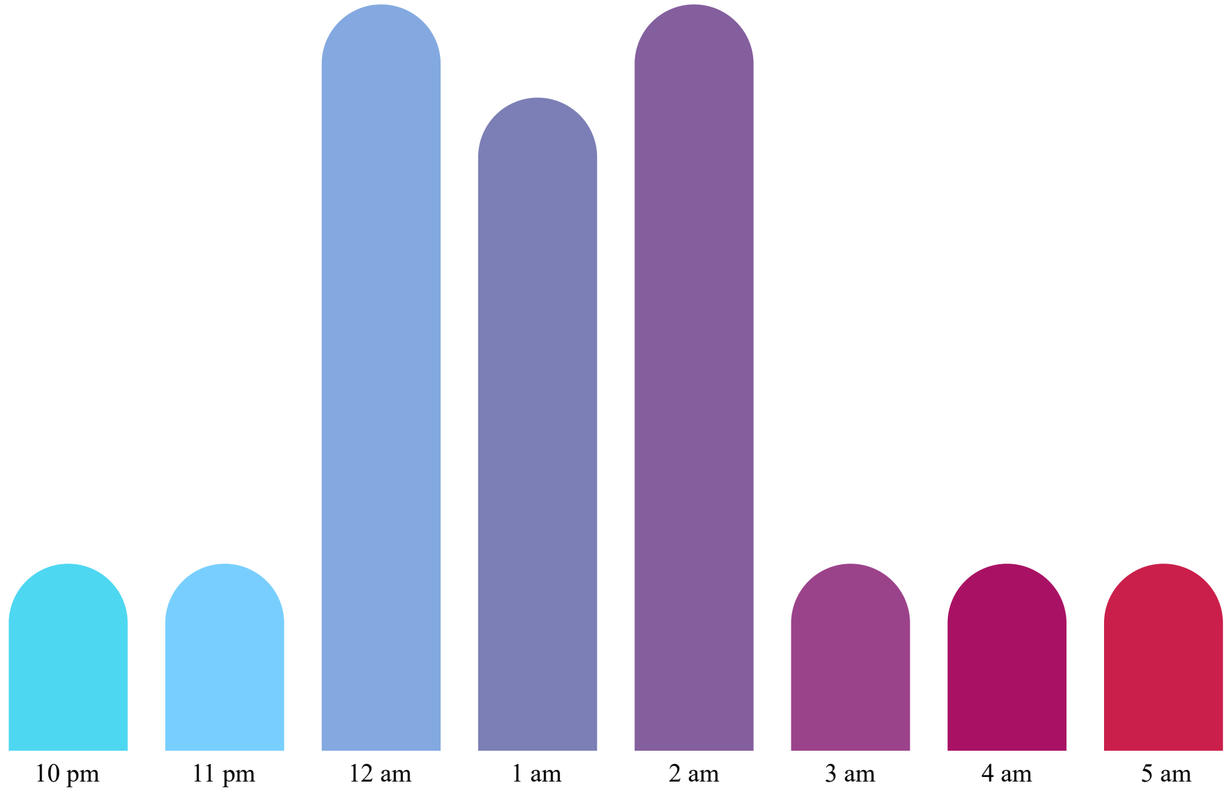


Lightning Sales or Flash Sales

The Lightning Sale (often called a Flash Sale) on platforms like Amazon and Flipkart is a limited-time promotional event where selected products are offered at significantly reduced prices for a short period or until the available stock is sold out.

Genral Occurance

12 AM → 1 AM → 2 AM



Why Our Solution?

Limitations of Existing Systems

- Poll price once every 24 hours
- Show only historical charts, no live graph
- Require account creation and login
- Accept standard URLs only, no short links
- Show ads and affiliate banners on dashboard

Our Unique Solutions

- Polls real website every 15 seconds via headless browser
- Shows two live graphs — real-time + full session history
- Zero login required, paste URL and start instantly
- Accepts any format — full URL, amzn.in, dl.flipkart.com, WhatsApp share text
- Zero ads, clean dashboard with live timer and poll counter

Literature Survey(1)

Sr. No.	Title of Paper	Author	Major Observations
1.	Dynamic Price Monitoring in E-Commerce: Leveraging Web Scraping, Data Analysis, and Email Automation for Dynamic Pricing Insights	A. A. Patel, A. R. Bhandare, A. S. Chauhan, S. V. Ghume, T. V. Rajarshi — PVPIT College Bavdhan, Pune	**Observations from the Paper** • This paper focuses on a web application that leverages web scraping, data storage, and email automation technologies to track real-time price changes for specific products on e-commerce platforms. It gathers key pricing metrics including the average, lowest, highest, and current prices, and presents this information visually to help users make informed purchasing decisions **Limitations Observed (Research Gap)** • the only difference is they used Next.js and MongoDB while we used React and Flask • Not specified (mentions regular scraping cycles only) https://doi.org/10.22214/ijraset.2024.64786

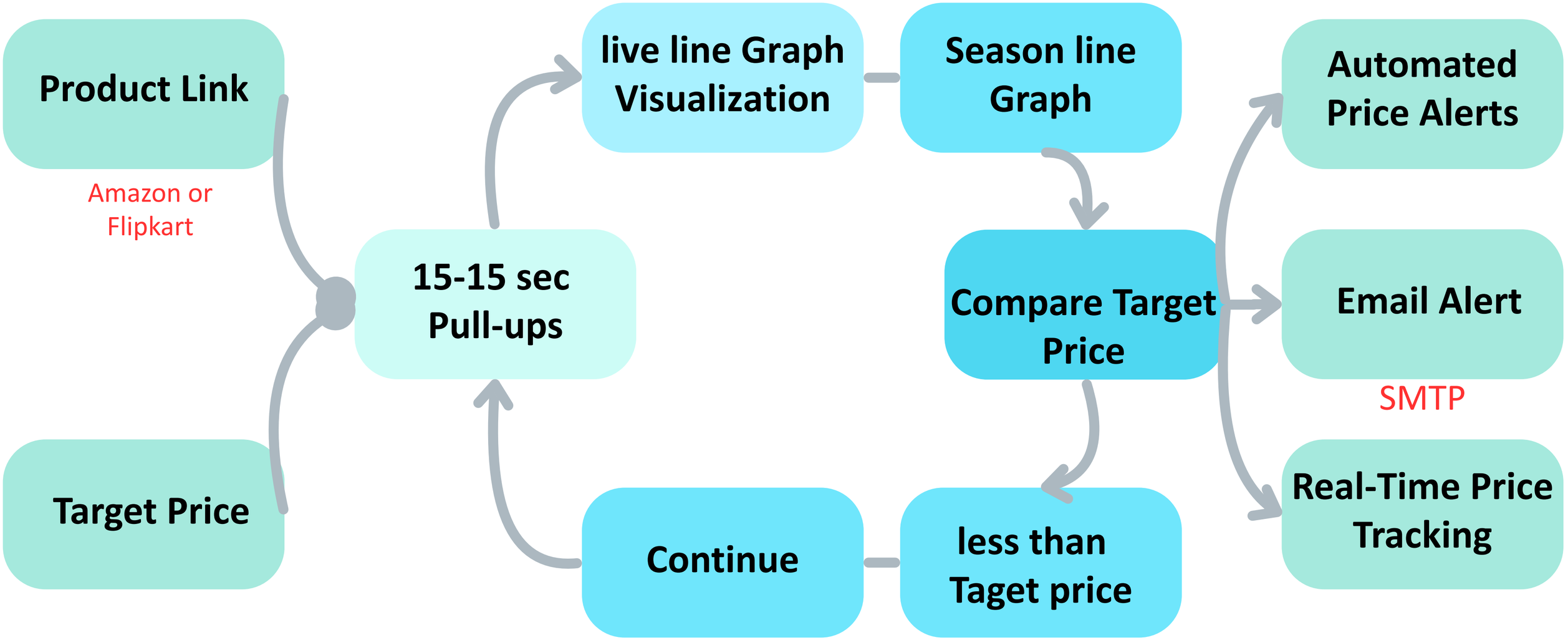
Literature Survey(2)

Sr. No.	Title of Paper	Author	Major Observations
2.	Online Shopping Item Cost Analysis through Web Scraping and Node.js	Authors: Dr. K. V. Nagendra, Dr. M. Ussenaiah — Sree Venkateswara College of Engineering, Nellore, AP	**Observation** The paper proposes a solution leveraging web scraping techniques to collect and analyze product prices from various e-commerce websites, with a user-friendly interface where consumers can input product details and receive comprehensive price comparisons across different online retailers. This paper directly validates the consumer-facing price comparison tool concept that PriceRadar implements. https://www.ijisrt.com/assets/upload/files/IJISRT24JUN1016.pdf

Literature Survey(3)

Sr. No.	Title of Paper	Author	Major Observations
3.	Research on Real-Time E-Commerce Price Comparison System Using Python Web Scraping Technology	F. Chen	**Observation** This paper presents a study on a real-time price comparison system using Python-based web scraping technologies, analyzing and comparing product prices across major e-commerce platforms by leveraging different web scraping libraries. It outlines the technical challenges associated with dynamic content extraction, anti-scraping mechanisms, and the effectiveness of various scraping tools, and enables users to monitor real-time price updates across multiple platforms. https://www.researchgate.net/publication/385269651_Research_on_Real-time_E-commerce_Price_Comparison_System_Using_Python_Web_Scraping_Technology

Methodology (Tentitavive Flow of Project)



PRICE RADAR

Platform	PRICE RADAR	Buyhatke	Price History	Camel Camel Camel
1.Price update speed	15 sec	24 hr	24 hr	24 hr
2.Live graph	✓	Historical only	Historical only	Historical only
3.Session history graph	✓	✗	✗	✗
4.Short URL support	✓	✗	✗	✗
5.WhatsApp share text input	✓	✗	✗	✗

SOFTWARE REQUIREMENTS

PriceRadar – Real-time E-commerce Price Tracking & Alert System

PriceRadar

1. FRONTEND DEVELOPMENT

- **React.js** – Interactive user interface development
- **JavaScript (ES6+)** – Application logic
- **HTML5 & CSS3** – Structure and styling
- **Recharts** – Real-time price visualization
- **Axios / Fetch API** – REST API communication

2. BACKEND DEVELOPMENT

- **Python 3.x** – Core backend programming language
- **Flask** – REST API framework
- **Flask-CORS** – Cross-Origin Resource Sharing
- **Playwright** – Browser automation and web scraping
- **smtplib (SMTP SSL)** – Automated email alerts

3. WEB SCRAPING & AUTOMATION

- **Playwright Chromium** – Headless browser automation
- Anti-bot browser configuration (stealth, headers, spoofing)
- Dynamic content rendering & JavaScript execution
- Resource blocking (images, fonts) for faster scraping

4. DEVELOPMENT TOOLS

- **Visual Studio Code** (VS Code) – Code editor
- **Node.js & npm** – React runtime and package management
- **Git & GitHub** – Version control and collaboration
- **Postman** – REST API testing

5. DEPLOYMENT PLATFORMS

- **Vercel** – React frontend hosting
- **Render** – Flask backend hosting

6. OPERATING SYSTEM

- **Windows 10/11** (Development)
- **Linux** / Cloud Server (Deployment)

7. BROWSER SUPPORT

- **Google Chrome** (Recommended)
- **Microsoft Edge**
- **Mozilla Firefox**

SOFTWARE ARCHITECTURE



References (Journal Papers,Website, Book)

[1] A. A. Patel, A. R. Bhandare, A. S. Chauhan, S. V. Ghume and T. V. Rajarshi,
"Dynamic Price Monitoring in E-Commerce: Leveraging Web Scraping, Data Analysis, and Email Automation for Dynamic Pricing Insights," International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 12, no. X, pp. 1225–, Oct. 2024.
DOI: 10.22214/ijraset.2024.64786

[2] K. V. Nagendra and M. Ussenaiah, "Online Shopping Item Cost Analysis through Web Scraping and Node.js," International Journal of Innovative Science and Research Technology (IJISRT), vol. 9, no. 6, pp. 338–, Jun. 2024.
DOI: 10.38124/ijisrt/IJISRT24JUN1016

[3] F. Chen, "Research on Real-time E-commerce Price Comparison System Using Python Web Scraping Technology," International Journal of Computer Science and Information Technology, vol. 16, no. 1, pp. 75–89, 2024.

Thank You